



DATABASE EXTREME
PRODUCT OVERVIEW · V1.3.0

A DATABASE FOR MULTI-TENANT PRODUCTS

Every customer gets a database.

Durable application state and vector memory, organized around the tenant boundary and operated from one self-hosted product.

01 Tenant-first 02 KV + vectors 03 Self-hosted

dbx · product experience

DBX 20:33:52 :6380

Product Docs Updates 1 Pricing Security Contact

GitHub Get started Search Ctr1+K

MEMORY ENGINE, NOT A SHARED CLUSTER.

One isolated engine per customer.

DBX holds working state and vector memory in the same process — and that process belongs to one tenant. Isolation is a directory, a WAL, and an HNSW index. Not a prefix you hope nobody forgets.

Run it locally

Watch the walkthrough

Isolated engines

RESP :6380

100 tenants / node

BSL 1.1

Browser sketch of the product, not a live node. GET on harbor cannot see acme's session.

ISOLATION · IN-BROWSER

acme harbor lumen

ACME

HARBOR

LUMEN

session	empty	session	empty	session	empty
vectors	0	vectors	0	vectors	0
WAL seq	0	WAL seq	0	WAL seq	0

acme> AUTH acme:writer ***

acme> AUTH harbor · SET session:1 hello · GET · VADD · VSIM

Send

V1.2.0 · SINGLE NODE

Supported profile: **100 tenants/node**, **100k vectors/tenant**, durable **strings + vectors**. Async WAL replicas optional (primary acks locally). Raft, cluster, and non-string RESP mutations **fail closed**.

13 SEP 2026 · VECTOR UPGRADES

VHIGRATE

keeps live search available while a shadow index is built.

VSEARCH ... AS_OF

replays retained WAL history for audit and RAG incident reproduction.

[Read the upgrade note →](#)

SET

GET

VADD

VSEARCH

VSIM

VFUSE

SPACE

SNAPSHOT

HIBERNATE

PROVISION

Database, operator experience, and deployment in one DBX product

DBX · Database product deck

01 / 12

Tenant separation is easy to get wrong.

When customer boundaries live in application conventions, every workflow must remember them.

01

Keys carry identity

Tenant IDs are repeated in key prefixes and query paths.

02

Access checks repeat

Each service must apply the right customer scope.

03

Lifecycle fragments

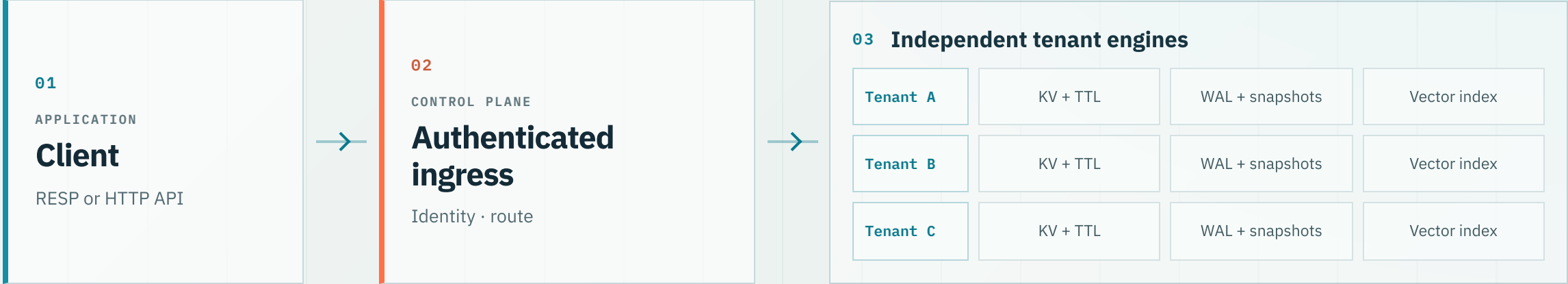
Backup and deletion span scripts and storage conventions.

DBX DESIGN CHOICE

Make the tenant the database boundary.

One ingress. *Independent tenant engines.*

Authenticate once, then route each request to its tenant-scoped data engine.



Working state and semantic recall, inside one tenant boundary.

01 STATE

Key-value store

Application state that is quick to read, update, and expire.

- ↳ RESP-compatible strings
- ↳ TTL, counters, and persistence
- ↳ Redis client libraries: Python, Node, Go

```
SET session:42 "active" EX 1800
```

+
ONE ENGINE

02 RECALL

Vector search

Caller-provided embeddings indexed for approximate nearest-neighbor retrieval.

- ↳ Sharded HNSW + SQ8 storage
- ↳ VSEARCH and VSIM
- ↳ Weighted vector fusion with VFUSE

```
VSEARCH docs query top_k 10
```

Embeddings come from the caller. DBX does not run embedding, image, or multimodal models.

Explore vector retrieval before wiring an app.

Try a tenant index, adjust search controls, and inspect ranked results from one workspace.

VSEARCH

VSIM · VFUSE

DBX
CONTROL PLANE

TENANT

- Overview
- Data Explorer
- Vector Playground
- Tenant keys
- Console

RUNTIME

- Hardware
- Storage
- Network
- Hosting

CONTROL PLANE

- Settings
- Security
- Replication

Tenant > searchlab > Vector searchlab

Search

tenant / searchlab / vector playground

Vector Playground

Query this tenant's vector index the same way an application would.

RUNNING

HTTP :8081

RESP :6401

Keys 6

Scores are cosine similarity × 100. VSIM reads the stored row and excludes that id. DBX does not run a model.

Similarity search

Semantic

Similar-to-Id

Multimodal

INDEX NAME

test_index

STORED VECTOR ID

Results

#1

36.69

payroll handbook and benefits

ID: bravo

#2

18.88

Vector Playground · query controls and ranked neighbors in the DBX dashboard

From live health to raw data, without leaving the product.

DBX

CONTROL PLANE

Tenant > Lumen Agent Platform > Overview > lumen-agents

Search

Backup

Overview

Data Explorer

Vector Playground

Console

Hardware

Storage

Network

Hosting

Settings

Security

Replication

Overview

Live command rate, memory, and keyspace for this tenant engine.

RUNNING

HTTP :8083

RESP :6403

Keys 21

OPERATIONS/SEC

0

total 60

MEMORY USED

2.32 MB

runtime allocation

ACTIVE CLIENTS

0

current connections

AVG LATENCY

1.82ms

command average

Throughput (ops/s)

No commands in the sampling window.

Key distribution

String

20

Vector

1

OVERVIEW · runtime health and key distribution

DBX

CONTROL PLANE

Tenant > Lumen Agent Platform > Console > lumen-agents

Search

Console

Issue RESP commands against this tenant's engine. Prompt shows the live RESP port.

RUNNING

HTTP :8083

RESP :6403

Keys 21

DBX console - Tenant: lumen-agents

Type a RESP command and press Enter (e.g. PING or SET mykey hello)

127.0.0.1:6403> GET ticket:T-1042

556

{ "status": "open", "priority": "high", "assignee": "copilot" }

CONSOLE · issue RESP commands

DBX

CONTROL PLANE

Tenant > Lumen Agent Platform > Explorer > lumen-agents

Search

Data Explorer

Browse keys in this tenant and inspect string values as JSON when possible.

New key

Refresh

RUNNING

HTTP :8083

RESP :6403

Keys 21

Filter keys...

KEY

TYPE

doc:big_web_index:faq-hours

STRING

doc:big_web_index:ticket-1120

STRING

session:acme-u-1042

STRING

ticket:T-1108

STRING

session:acme-u-2014

STRING

doc:big_web_index:faq-sla

STRING

session:acme-u-1901

STRING

agent:router:config

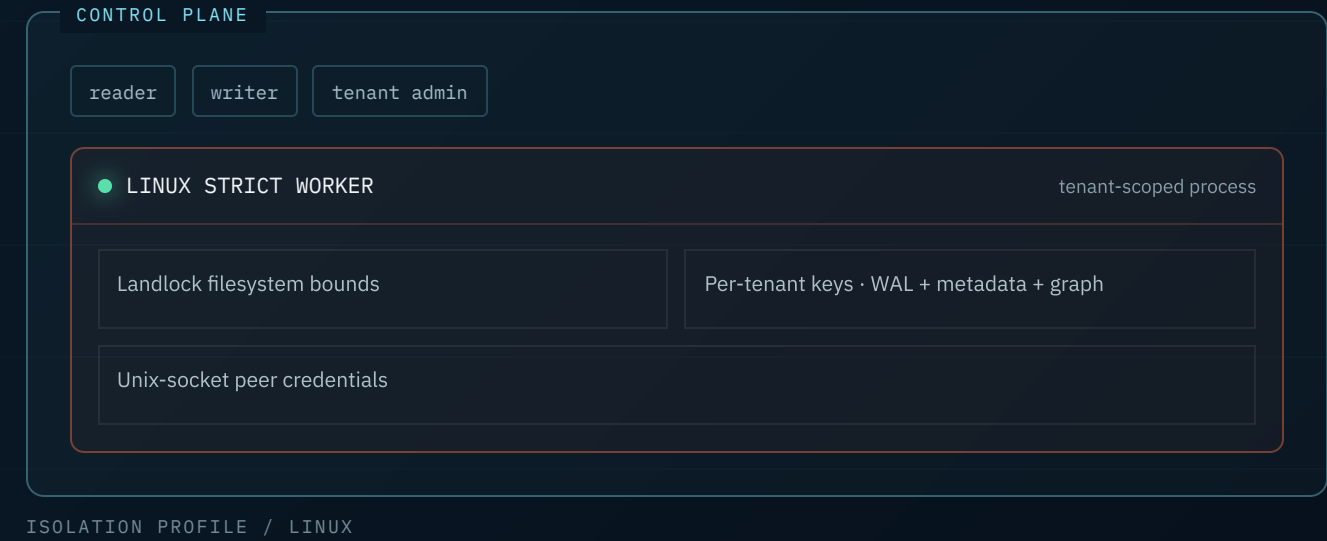
STRING

agent:scratch-u-1901

STRING

DATA EXPLORER · inspect tenant keys

Tenant separation reaches the worker and its data lifecycle.



14–17 MiB idle RSS per strict worker

Clear separation. Honest storage boundary.

Scoped roles, quotas, usage APIs, and tenant lifecycle actions are managed through the control plane.

AT-REST DETAIL

Vector payload rows are mmap-backed and not encrypted by DBX. Use encrypted storage such as LUKS or fscrypt when at-rest confidentiality is required. Data is plaintext inside a running worker.

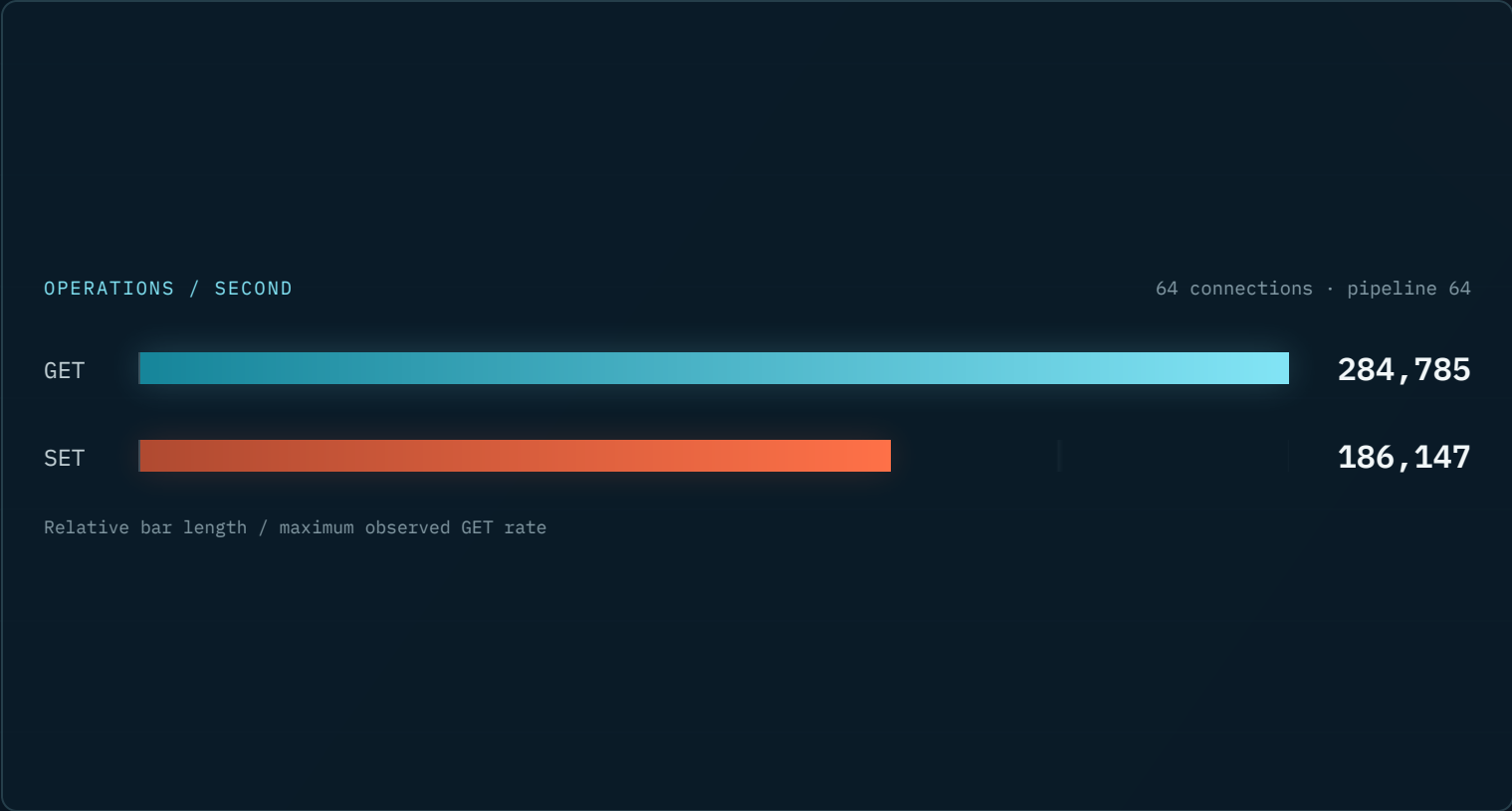
Customer-level operations are built into the data model.

01	Recover Checksummed WAL and checkpoints restore tenant data after restart.	WAL + RECOVERY
03	Change safely Build and promote a replacement vector index while the current index remains searchable.	SHADOW MIGRATION
05	Replicate Optional replicas stream authenticated, encrypted WAL; primary writes acknowledge locally.	ENCRYPTED WAL

02	Back up Versioned tenant archives carry SHA-256 checksums across KV and vector state.	TENANT ARCHIVE
04	Investigate Search retained vector state as of a timestamp for retrieval analysis.	TIME-TRAVEL
06	Support Scoped diagnostics and opt-in audited wake accompany the embedded operator dashboard.	SUPPORT + UI

Fast key-value operations on a measured host.

CERTIFICATION RUN
27 AUG 2026



DBX / KV

128k operations per workload

1 tenant · no eviction

Every second WAL sync setting

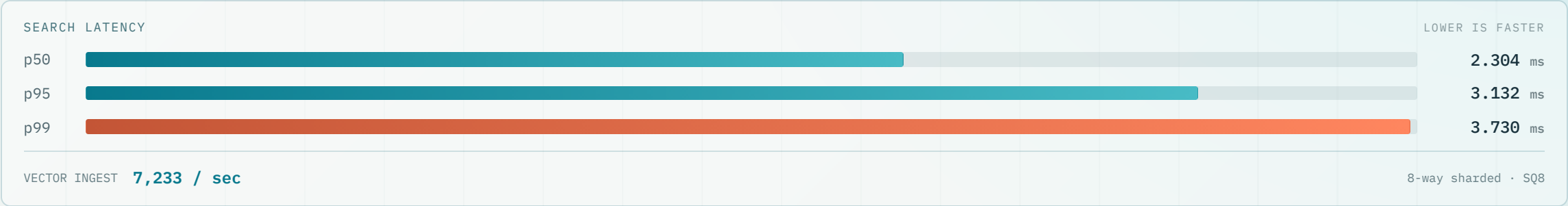
Windows 10 · 18 logical processors · 15.6 GiB RAM
Loopback RESP · 27 Aug 2026

Measured on the Windows 10 certification host; this run does not certify Linux performance. Results vary with host, durability settings, workload, and client. Methodology: [scripts/benchmarks/performance_analysis.md](#).

Latency with recall, on the same slide.

100,000 vectors × 128 dimensions · float32 brute-force recall baseline

RECALL@10 MEAN
0.920 p05 · 0.800



27 Aug 2026 · Windows 10 · 18 logical processors · 15.6 GiB RAM. Linux CI reruns vector gates; this Windows host does not certify Linux performance. Methodology: [scripts/benchmarks/performance_analysis.md](#).

Built for bounded, per-customer working sets.

A self-hosted database with tenant-scoped application state, vector search, and an included operator dashboard.

100

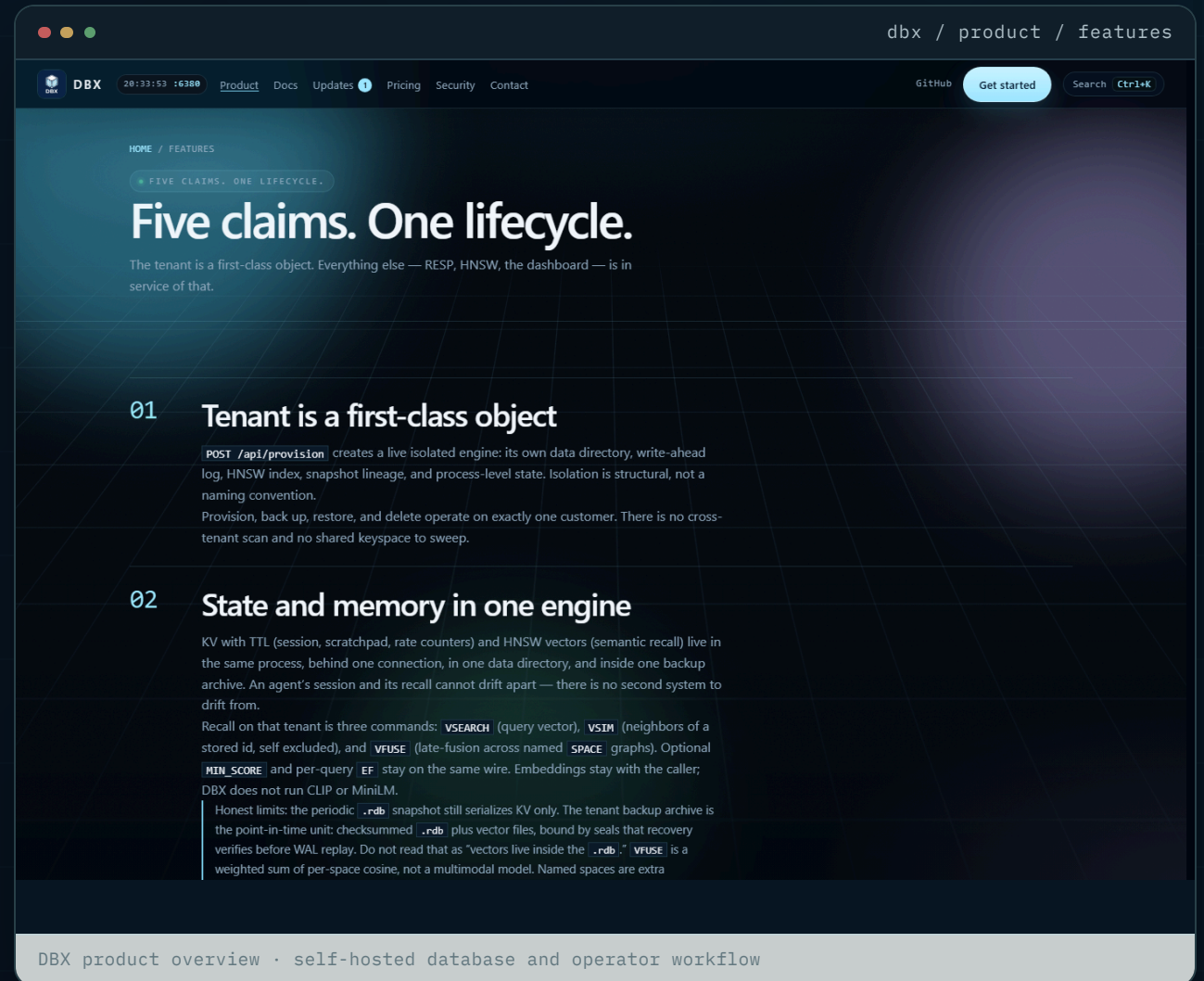
tenants / node

100k

vectors / tenant

CURRENT BOUNDARIES

Single-node v1 · no cluster or cross-node sharding · no SQL joins · no billion-vector claim.





DATABASE EXTREME

DBX · V1.3.0

A DATABASE BOUNDARY FOR EVERY CUSTOMER

One customer. One database.

Durable state, vector memory, and tenant-level operations in one self-hosted database.

dbxdb.co.in

github.com/vanshjain-0702/DBX-Database-Extreme

